Programs on String and String Buffer classes

1. Write a Java program to check whether a given string is a palindrome using the String class.

```
public class PalindromeCheck {
    public static void main(String[] args) {
        String str = "madam";
        String rev = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            rev = rev + str.charAt(i);
        }

        if (str.equals(rev)) {
            System.out.println(str + " is a palindrome.");
        } else {
            System.out.println(str + " is not a palindrome.");
        }
    }
}
```

2. Write a Java program to count the number of vowels, consonants, digits, and special characters in a string.

```
public class CountCharacters {
    public static void main(String[] args) {
        String str = "Hello123@Java!";
        int vowels = 0, consonants = 0, digits = 0, special = 0;

        str = str.toLowerCase();
        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);

            if (ch >= 'a' && ch <= 'z') {
                if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
                    vowels++;
                else
                    consonants++;
            } else if (ch >= '0' && ch <= '9') {
                digits++;
            } else {
                special++;
            }
        }

        System.out.println("Vowels: " + vowels);
        System.out.println("Consonants: " + consonants);
        System.out.println("Digits: " + digits);
        System.out.println("Special Characters: " + special);
    }
}
```

3. Write a Java program to compare two strings lexicographically without using the compareTo() method.

```java
public class LexicographicCompare {
    public static void main(String[] args) {
        String str1 = "apple";
        String str2 = "apricot";

        int minLength = Math.min(str1.length(), str2.length());
        boolean equal = true;

        for (int i = 0; i < minLength; i++) {
            if (str1.charAt(i) != str2.charAt(i)) {
                if (str1.charAt(i) < str2.charAt(i)) {
                    System.out.println(str1 + " comes before " + str2);
                } else {
                    System.out.println(str2 + " comes before " + str1);
                }
                equal = false;
                break;
            }
        }
        if (equal) {
            if (str1.length() == str2.length()) {
                System.out.println("Both strings are equal.");
            } else if (str1.length() < str2.length()) {
                System.out.println(str1 + " comes before " + str2);
            } else {
                System.out.println(str2 + " comes before " + str1);
            }
        }
    }
}
```

4. Write a Java program that uses StringBuffer to reverse a string and display the output.

```
public class ReverseStringBuffer {
    public static void main(String[] args) {
        String str = "Programming";
        StringBuffer sb = new StringBuffer(str);

        System.out.println("Original String: " + str);
        System.out.println("Reversed String: " + sb.reverse());
    }
}
```

5. Write a Java program using StringBuffer to insert, delete, and replace characters in a given string.

```
public class StringBufferOperations {
    public static void main(String[] args) {
        StringBuffer sb = new StringBuffer("Hello World");

        // Insert
        sb.insert(5, " Java");
        System.out.println("After Insert: " + sb);

        // Delete
        sb.delete(5, 10);
        System.out.println("After Delete: " + sb);

        // Replace
        sb.replace(6, 11, "Universe");
        System.out.println("After Replace: " + sb);
    }
}
```